RAMLA IJAZ

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EDUCATION

Yale University — 2021 – 2026 (expected)
Ph.D., Computer Science — GPA: 4.0 / 4.0

Rice University — 2017 – 2020
M.S., Electrical and Computer Engineering — GPA: 3.66 / 4.0

University of Engineering and Technology, Lahore, Pakistan — 2012 – 2016
B.Sc., Electrical Engineering — ranked 2nd of 225 students — GPA: 3.87 / 4.0

CURRENT RESEARCH

Yale Efficient Computing Lab — New Haven, CT
Graduate Researcher — Jan 2021 – present

- Designing an AI agentic system that uses LLMs to transform Rust systems code, addressing the limitations of automatic C-to-Rust converters by restructuring code to minimize unsafe constructs and maximize compiler-enforced invariants. In the first phase, I use GPT-4o to generate and refine Hardware Abstraction Layers (HALs) that ensure datasheet-compliant device communication, alongside supporting tooling for working with LLMs.
- Developed a design methodology that combines type checking (a lightweight formal method) with formal verification to uphold system invariants. By leveraging Rust's affine types to prevent aliasing and integrating verification tools such as Prusti and Verus, I show how these techniques jointly enforce system guarantees with less overhead than full formal verification. I applied the methodology to the memory subsystem and the 10 GbE network driver in Theseus OS, achieving correctness without performance loss.

RESEARCH EXPERIENCE

Rice University — Houston, TX
M.S. Research — Jan 2018 – Dec 2020

- Rethought operating system design so that critical OS invariants are enforced directly by the Rust compiler. Applied this approach to key components of Theseus OS—inter-process communication (IPC), the heap allocator, network drivers, and power management—demonstrating stronger correctness guarantees without sacrificing performance.

Microsoft Research, Cambridge (UK) — Virtual
Research Intern — host: Bozidar Radunovic — Jun 2020 – Aug 2020

- Conducted a detailed performance analysis of each stage of the software-based RAN baseband processing pipeline on CPUs and GPUs, exploring the offloading of LDPC decoding to accelerators (GPUs, FPGAs, eASICs) to optimize processing efficiency.
- Built a simulator that intelligently partitions baseband processing and inference workloads across machines in an edge data center, finding configurations that optimize resource utilization while meeting the deadlines of low-latency applications.

Electrical Engineering Department, LUMS Lahore, Pakistan
Research Assistant — Dr. Adeel Pasha & Dr. Naveed ul Hassan Jan 2017 – Jun 2017

- Contributed to the National ICT R&D project "Indoor Location-Based Services (LBS) for mobile devices."

Electronics and Embedded Systems Lab, LUMS Lahore, Pakistan
Research Assistant Jul 2016 – Dec 2016

- Researched lightweight encryption for the Internet of Things and proposed a hardware implementation of the TAV-128 hash algorithm that improves security while maintaining low power and resource use.

Network and Systems Group, LUMS Lahore, Pakistan
Research Intern — host: Dr. Ihsan Ayub Qazi Jun 2015 – Sep 2015

- Developed algorithms for routing in wireless data-center environments.

PUBLICATIONS

Ramla Ijaz, Kevin Boos, and Lin Zhong. "Leveraging Rust for Lightweight OS Correctness." *1st Workshop on Kernel Isolation, Safety and Verification (KISV)*, 2023.

Ramla Ijaz. "Exploring Intralingual Design in Operating Systems." M.S. Thesis, Rice University, 2020.

Kevin Boos, Namitha Liyanage, Ramla Ijaz, and Lin Zhong. "Theseus: an Experiment in Operating System Structure and State Management." *14th USENIX Symposium on Operating Systems Design and Implementation (OSDI)*, 2020.

Ramla Ijaz and Muhammad Adeel Pasha. "Area-Efficient and High-Throughput Hardware Implementations of the TAV-128 Hash Function for Resource-Constrained IoT Devices." *IEEE IDAACS*, 2017.

TEACHING

CPSC 429 — guest lecture: Correctness by Combining Types and Verification Nov 2024

CPSC 429 — presentation: Rust Verification Tools + Demo Nov 2023

CPSC 323 — Introduction to Systems Programming, Teaching Fellow Jan – May 2022

CPSC 433 — Computer Networks, Teaching Fellow Jan – May 2021

SERVICE & LEADERSHIP

Yale Graduate Society of Women Engineers — Mentorship Chair Sep 2024 – May 2025
Created mentorship families for SWE members and organized events.

CS Graduate Student Advisory Committee — Board Member Jun 2022 – May 2023
Organized events for CS graduate students and worked with faculty to improve the graduate experience.

Yale Pathways for Science — Volunteer Summer 2021
Created assignments for a cloud-based chat application to introduce high school students to the cloud.

AWARDS & SCHOLARSHIPS

- NSDI '21 Diversity Grant recipient
- MobiCom '19 N2Women Student Travel Grant recipient

- OSDI ’18 Student Travel Grant recipient
- President’s Prize Fellowship, Rice University (2017)
- Texas Instruments Distinguished Fellowship (2017)
- Graduated with honors, B.Sc. Electrical Engineering (2016)

SKILLS

Programming languages: Rust, C, C++, Python, CUDA, Dafny, Java, Verilog
Tools & frameworks: OpenAI API, Prusti, Verus, rustc, Linux, MATLAB